Ritam Ghosh

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GitHub | LinkedIn | Codeforces / TheOrion

EXPERIENCE

Antahin Abha Foundation | Kolkata, WB, India

Technology Intern | June 2026 - Present

- Building the **AskAbha Chatbot v2** end-to-end, architecting a RAG engine that retrieves and ranks context by document metadata to ground responses and improve relevance.

PROJECTS

Ask the Book Hybrid RAG System

Python, LlamaIndex, ChromaDB, Groq, Streamlit | GitHub | Live demo

- Built a hybrid RAG pipeline fusing semantic search and **BM25** via Reciprocal Rank Fusion, re-scored by a **Cross-Encoder** for precise QA.
- Achieved **90% accuracy** on a custom evaluation suite with **sub-3s latency**; grounded responses with automatic citations to prevent hallucinations.
- Implemented LlamaIndex semantic chunking and streamed real-time responses using **Groq Llama 3.1 8B**.

AlgoCoach ML Problem Recommender

React, FastAPI, scikit-learn, httpx | GitHub | Live demo

- Developed a KNN-based engine analyzing Codeforces submissions to identify and target algorithmic weaknesses.
- Engineered a feature vectorization scheme weighting tag relevance over rating to power **Top-5** and **Staircase** recommendation modes.
- Built an async FastAPI backend with concurrent API fetching and a disk-cached database, deployed across Render and Vercel.

EDUCATION

Jadavpur University | 2024 - 2028

Bachelor of Computer Science and Engineering | CGPA: **8.0 / 10.0**

ACHIEVEMENTS

- **Competitive programming:** Expert on Codeforces (max: 1732), 4-Star on CodeChef (max: 1816), Knight on LeetCode (2071). Solved **2000+** problems across Codeforces, LeetCode, CodeChef, and AtCoder.
- **Global rank 271** out of 15,000+ participants in Codeforces Round 1089 (Div. 2).
- **Global rank 364** out of 15,000+ participants in Codeforces Educational Round 185.
- **Winner (Rank 1), Epochalypse at Srijan 2026:** engineered an automated Market Maker algorithm to achieve optimal profit through real-time liquidity provision.
- **Winner (Rank 1), Uncode at Srijan 2026, Jadavpur University:** solved reverse-engineered algorithmic challenges based on input-output pattern recognition.
- **Competitive examinations:** JEE Advanced rank 9102, JEE Mains 98.8 percentile, WBJEE state rank 191.

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript/TypeScript, Java, SQL, HTML/CSS

Frameworks & libraries: React.js, Node.js, FastAPI, scikit-learn, LlamaIndex, Streamlit, Tailwind CSS, WebSockets

Tools & infrastructure: Git/GitHub, Docker, Redis (Upstash), ChromaDB, Groq API, Jupyter, Google Colab